

Ethical Frameworks in Applied Data Science: VSD vs. DPIA

Selected Ethical Models:

In the field of applied data science, two important ethical models are Value Sensitive Design (VSD) and the Data Protection Impact Assessment (DPIA). VSD is a design method that puts human values, like fairness and transparency, directly into the engineering process (Friedman & Hendry, 2019). DPIA is a systematic process required by the GDPR. It is used to find privacy risks and make them smaller in data systems before they are deployed (European Union, 2016).

Similarities and Differences

Both models are proactive. This means they are used to find and stop ethical problems before a system goes into production. They both need you to consider all relevant stakeholders and try to prevent harm to the people whose data is used.

The biggest difference is in their scope and how they are used in the machine learning lifecycle. VSD is a broad approach that directly influences the technical choices we make, for example, fixing algorithmic bias during feature engineering or choosing models that are easy to explain. DPIA is mostly about legal rules and risk management. It focuses only on data governance, checking if data collection is secure and legally justified. It does not really consider the mathematical models we apply to the data.

Personal Preference

For my work in applied data science, my preference goes to VSD. Even though legal tools like the DPIA are mandatory, they can cause a checkbox mentality where people just fill in a form without really thinking about the stakeholders. VSD forces data scientists to translate abstract moral concepts into technical metrics. Because we put ethics into the optimization of machine learning models, VSD helps us to actively build fair systems instead of only looking at the law afterwards.

Most Suited for Data Ethics

Even though VSD is better for designing fair algorithms, I think DPIA is the most suitable framework for the data ethics part in applied data science. Our work is based on collecting and processing very large datasets. Because of this, data ethics is mostly about data minimization, informed consent, privacy rights, and lawful processing. DPIA gives us a clear and standard method to check our data pipelines. It forces the organization to map where the data comes from and where it goes, explain why we need to keep the data, and verify the security measures. For these reasons, DPIA is a very practical tool to make sure the raw data we use for our predictive models is handled in an ethical way.

References

European Union. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Official Journal of the European Union, L 119, 1-88.

Friedman, B., & Hendry, D. G. (2019). Value Sensitive Design: Shaping Technology with Moral Imagination. MIT Press.